

User Manual

for S32G LIN_LLCE Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	29.09.2023	NXP LLCE Team	Prepared for release S32G LLCE AUTOSAR 4.4 Version 1.0.7

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductor AUTOSAR Lin_43_LLCE for S32G. AUTOSAR Lin_43_LLCE driver configuration parameters and deviations from the specification are described in Driver chapter of this document. AUTOSAR Lin_43_LLCE driver requirements and APIs are described in the AUTOSAR Lin_43_LLCE driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Development Error Tracer
ECUM	ECU state Manager
ECU	Electronic Control Unit
ISR	Interrupt Service Routine
LIN	Local Interconnect Network
LSB	Least Significant Bit
MCU	Micro Controller Unit
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of LIN Driver	AUTOSAR Release 4.4.0
2	S32G2 Reference Manual	Rev. 5, 05/2022
3	S32G2 Data Sheet	Rev. 5, 05/2022
4	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 0P77B, Rev. 3.1
5	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 1P77B, Rev. 1.1
6	S32G3 Reference Manual	Rev. 2, 09/2022
7	S32G3 Data Sheet	Rev. 2, Draft C, 09/2021
8	S32G3 Errata Document	S32G3 Mask Set Errata for Mask 1P72B, Rev. 2.2

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)

3.1 Requirements

Requirements for this driver are detailed in the Autosar Driver Software Specification document (See [Table Reference List](#)).

3.2 Driver Design Summary

The LIN driver is part of the microcontroller abstraction layer (MCAL), performs the hardware access and offers a hardware independent API to the upper layer.

The only upper layer, which has access to the LIN driver, is the LIN Interface.

A LIN driver can support more than one channel.

This means that the LIN driver can handle one or more LIN channels as long as they belong to the same LIN hardware unit.

The LIN Driver for S32G, uses the LLCE module which provides special support for the LIN protocol.

It can be used to automate most tasks of a LIN master.

It is possible to transmit entire frames (or sequences of frames) and receive data from LIN slaves without any CPU intervention.

The LIN physical interface should be connected to the LLCE LIN module pins in order to get the LIN bus voltage levels.

The S32G contains up to 4 blocks.

The LinLLCE has the following major features:

- LIN Master Node functionality (master and slave task)
- LIN Slave Node functionality (slave task)
- Compatible with LIN from revisions 1.x and 2.x of the LIN standard
- Detection of Bit Errors, Physical Bus Errors and Checksum Errors
- All status bit can generate maskable interrupts
- Application layer CRC support
- Programmable CRC polynom
- Double Stop Flag insertion after Bit Errors
- Detection and generation of wakeup characters
- Programmable wakeup delimiter time
- Can be configured to include header bits in checksum

3.3 Hardware Resources

The device family includes S32G274A derivatives.

The hardware configured by the Lin driver is LLCE.

Lin Physical Channel (4 channels):** LINFlexD_0, LINFlexD_1, LINFlexD_2, LINFlexD_3.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR *driver* Driver software specification in some places. The table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the *driver* Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the driver.

Requirement	Status	Description	Notes
SWS_Lin_00055	N/S	The Lin module shall fulfill all design and implementation guidelines as described in Specification of C Implementation Rules AUTOSAR_TR_C↔ImplementationRules.pdf.	Requirement already covered by process.
SWS_Lin_00026	N/S	If the LIN hardware unit cannot queue the bytes for transmission or reception (e.g. simple UART implementation), the LIN driver shall provide a temporary communication buffer.	The LIN hardware already has a data buffer built-in.
SWS_Lin_00039	N/S	Values that can be configured are hardware dependent. Therefore, the rules and constraints cannot be given in the standard.	This is not a requirement.

Requirement	Status	Description	Notes
SWS_Lin_00999	N/S	These requirements are not applicable to this specification. (SRS_BSW_00307, SRS_BSW_00312, SRS_BSW_00325, SRS_BSW_00326, SRS_BSW_00328, SRS_BSW_00329, SRS_BSW_00330, SRS_BSW_00331, SRS_BSW_00336, SRS_BSW_00339, SRS_BSW_00342, SRS_BSW_00343, SRS_BSW_00353, SRS_BSW_00357, SRS_BSW_00359, SRS_BSW_00360, SRS_BSW_00361, SRS_BSW_00373, SRS_BSW_00376, SRS_BSW_00378, SRS_BSW_00383, SRS_BSW_00395, SRS_BSW_00397, SRS_BSW_00398, SRS_BSW_00399, SRS_BSW_00400, SRS_BSW_00413, SRS_BSW_00415, SRS_BSW_00416, SRS_BSW_00417, BSW00420, SRS_BSW_00422, SRS_BSW_00423, SRS_BSW_00424, SRS_BSW_00425, SRS_BSW_00426, SRS_BSW_00427, SRS_BSW_00428, SRS_BSW_00429, BSW00431, SRS_BSW_00432, SRS_BSW_00433, BSW00434, SRS_BSW_00005, SRS_BSW_00007, SRS_BSW_00162, SRS_BSW_00168, SRS_SPAL_12056, SRS_SPAL_12267, SRS_SPAL_12163, SRS_SPAL_12463, SRS_SPAL_12075, SRS_SPAL_12078, SRS_SPAL_12092, SRS_Lin_01551, SRS_Lin_01568, SRS_Lin_01569, SRS_Lin_01570, SRS_Lin_01564, SRS_Lin_01546, SRS_Lin_01561, SRS_Lin_01549, SRS_Lin_01571, SRS_Lin_01514, SRS_Lin_01515, SRS_Lin_01502, SRS_Lin_01558, BSW01527, SRS_Lin_01523, SRS_Lin_01540, SRS_Lin_01545, SRS_Lin_01534, SRS_Lin_01574, SRS_Lin_01539, SRS_Lin_01544, SRS_Lin_01590).	This is not a requirement.

3.5 Driver Limitations

None.

3.6 Driver usage and configuration tips

3.6.1 Frame Timeout Disable Feature

The 'Lin Channel Disable Frame Timeout' checkbox in LinChannel container should be enabled in order to have this

Driver

feature active. If LinChannelDisableFrameTimeout is ON then Lin driver will accept the frame that is longer than Maximal Frame Length. For the LIN driver, timeout feature is applied to both LIN 2.1 master and slave nodes, in response frame reception.

Note

In slave mode, this feature must be set to false to avoid errata e7274

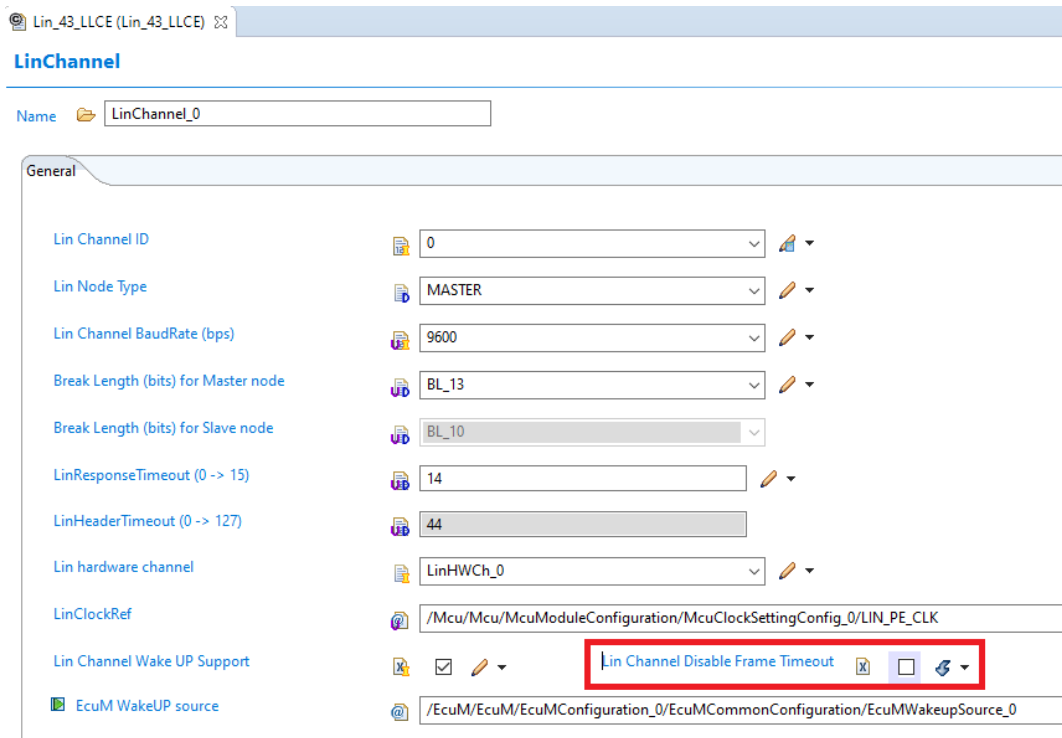


Figure 3.1 Frame Timeout Disable Feature configuration node in EB Tresos.

3.7 Runtime errors

The Lin driver generates the following DEM errors at runtime.

Function	Error Code	Condition triggering the error
Lin_43_LLCE_GoToSleep()	Lin_E_TIMEOUT	Timeout caused by hardware error waiting for cancellation of current byte. No sleep command will be sent , and Lin driver will not enter sleep state.
Lin_43_LLCE_GoToSleepInternal()	Lin_E_TIMEOUT	Timeout caused by hardware error waiting for cancellation of current byte. Lin driver will not enter sleep state.
Lin_43_LLCE_SendFrame()	Lin_E_TIMEOUT	Timeout caused by hardware error waiting for cancellation of current byte. New frame is not sent.

3.8 Symbolic Names Disclaimer

All containers having `symbolicNameValue` set to `TRUE` in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Lin](#)
 - Container [AutosarExt](#)
 - * Parameter [LinDisableDemReportErrorStatus](#)
 - * Parameter [LinEnableUserModeSupport](#)
 - Container [LinGeneral](#)
 - * Parameter [LinDevErrorDetect](#)
 - * Parameter [LinIndex](#)
 - * Parameter [LinTimeoutDuration](#)
 - * Parameter [LinVersionInfoApi](#)
 - * Reference [LinEcucPartitionRef](#)
 - Container [LinDemEventParameterRefs](#)
 - * Reference [LIN_E_TIMEOUT](#)
 - Container [LinGlobalConfig](#)
 - * Container [LinChannel](#)
 - Parameter [LinChannelId](#)
 - Parameter [LinNodeType](#)
 - Parameter [LinChannelBaudRate](#)
 - Parameter [BreakLengthMaster](#)
 - Parameter [BreakLengthSlave](#)
 - Parameter [LinResponseTimeout](#)
 - Parameter [LinHeaderTimeout](#)
 - Parameter [LinHwChannel](#)
 - Parameter [LinChannelWakeupSupport](#)
 - Parameter [LinChannelDisableFrameTimeout](#)
 - Reference [LinClockRef](#)
 - Reference [LinChannelEcuMWakeupSource](#)
 - Reference [LinChannelEcucPartitionRef](#)
 - Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)

- * Parameter [ArReleaseMinorVersion](#)
- * Parameter [ArReleaseRevisionVersion](#)
- * Parameter [ModuleId](#)
- * Parameter [SwMajorVersion](#)
- * Parameter [SwMinorVersion](#)
- * Parameter [SwPatchVersion](#)
- * Parameter [VendorApiInfix](#)
- * Parameter [VendorId](#)

4.1 Module Lin

Configuration of the Lin (Local Interconnect Network) module.

Included containers:

- [AutosarExt](#)
- [LinGeneral](#)
- [LinDemEventParameterRefs](#)
- [LinGlobalConfig](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container AutosarExt

This container contains the global configuration parameters of the Extended Autosar Lin driver.

This container is a `MultipleConfigurationContainer`, i.e. this container and its sub-containers exist once per configuration set.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter LinDisableDemReportErrorStatus

Switches the Diagnostic Error Reporting and Notification OFF.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.4 Parameter LinEnableUserModeSupport

When this parameter is enabled, the LIN module will adapt to run from User Mode

Note:

Lin module does not include registers protection. So, It is accessible to all registered in any public mode. Lin module is not affected by this field

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.5 Container LinGeneral

This container contains the parameters related to each LIN Driver Unit.

This container is a MultipleConfigurationContainer, i.e. this container and

its sub-containers exist once per configuration set.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.6 Parameter LinDevErrorDetect

Switches the Default Error Detection and Notification ON or OFF.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.7 Parameter LinIndex

Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

Note

This parameter is not used in the current implementation.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.8 Parameter LinTimeoutDuration

Specifies the maximum number of loops for blocking function until a timeout is raised in short term wait loops.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	0

4.9 Parameter LinVersionInfoApi

Switches the Lin_GetVersionInfo function ON or OFF.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.10 Reference LinEcucPartitionRef

Maps the Lin driver to zero or multiple ECUC partitions to make the modules API available in this partition.

The Lin driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.11 Container LinDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API

Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.12 Reference LIN_E_TIMEOUT

Reference to the DemEventParameter which shall be issued when the error

"Timeout caused by hardware error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.13 Container LinGlobalConfig

This container contains the global configuration parameter of the Lin driver.

This container is a `MultipleConfigurationContainer`, i.e. this container and its sub-containers exit once per configuration set.

Included subcontainers:

- [LinChannel](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.14 Container `LinChannel`

This container contains the configuration (parameters) of the LIN Controller(s).

Note:

"User should use unique names for naming the LIN channels across different `LinGlobalConfig` Sets."

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.15 Parameter `LinChannelId`

Identifies the LIN channel. Replaces `LIN_CHANNEL_INDEX_NAME` from the LIN SWS.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	65535
min	0

4.16 Parameter LinNodeType

Specifies the LIN node type of this channel: Master or Slave node

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	MASTER
literals	['MASTER', 'SLAVE']

4.17 Parameter LinChannelBaudRate

Specifies the baud rate of the LIN channel in 'bps'. Valid range: 4800..20000.

Note: The minimum value of the baudrate is 4800bps (AUTOSAR Standard: 1000bps), this information was specified in data sheet

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	9600
max	20000
min	1000

4.18 Parameter BreakLengthMaster

Defines the break length in bits for Master node.

Note:

This Parameter is an Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	BL_13
literals	['BL_13', 'BL_14', 'BL_15', 'BL_16', 'BL_17', 'BL_18', 'BL_19', 'BL_20', 'BL_21', 'BL_22', 'BL_23', 'BL_36', 'BL_50']

4.19 Parameter BreakLengthSlave

Defines the break length in bits which can detect by Slave node.

Note:



Tresos Configuration Plug-in

This Parameter is an Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	BL_10
literals	['BL_10', 'BL_11']

4.20 Parameter LinResponseTimeout

Response timeout value.

LINTOCR[RTO] will be configured by this node.

This is the response timeout duration (in bit time) for 1 byte.

The reset value is 0Eh = 14, corresponding to $T_Response_Maximum = 1.4 \times T_Response_Nominal$.

This node is only configured if LinChannelDisableFrameTimeout is false.

Note:

This Parameter is an Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	14
max	15
min	0

4.21 Parameter LinHeaderTimeout

Header timeout value

LINTOCR[HTO] will be configured by this node.

This field contains the header timeout duration (in bit time).

For slave, these bits should be programmed without considering 11 bits of break.

Please see sub-chapter "Timeout error" in Reference Manual to know how to calculate this value.

This node is only configured if LinChannelDisableFrameTimeout is false and LinNodeType is SLAVE.

Note:

This Parameter is an Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	44
max	127
min	0

4.22 Parameter LinHwChannel

Selects the Lin Hardware Channel.

Note:

This Parameter is an Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	LinHWCh_0
literals	['LinHWCh_0', 'LinHWCh_1', 'LinHWCh_2', 'LinHWCh_3']

4.23 Parameter LinChannelWakeupSupport

Specifies if the LIN hardware channel supports wake up functionality.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.24 Parameter LinChannelDisableFrameTimeout

When LinDisableFrameTimeout is ON, LIN driver will accept the frame that is longer than Maximal Frame Length.

Note

According to Lin protocol 2.0/2.1, the driver shall not check Timeout event. Consequently, Short Response and No Response errors will not be supported by LIN driver.

To avoid Errata ERR007274 this feature will be only applied for Master mode, Timeout will be always used by Slave mode and not depend on this node.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.25 Reference LinClockRef

Reference to the LIN clock source configuration, which is set in the MCU driver configuration.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSetting↔ Config/McuClockReferencePoint

4.26 Reference LinChannelEcuMWakeupSource

This parameter contains a reference to the Wakeup Source for this controller as defined in the ECU State Manager.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuM/EcuMConfiguration/EcuMCommon← Configuration/EcuMWakeupSource

4.27 Reference LinChannelEcucPartitionRef

Maps one single Lin channel to zero or one ECUC partitions.

The ECUC partition referenced is a subset of the ECUC partitions where the Lin driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcuPartitionCollection/EcuPartition

4.28 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.29 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.30 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

Property	Value
defaultValue	4
max	4
min	4

4.31 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.32 Parameter ModuleId

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	82
max	82
min	82

4.33 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	1
max	1
min	1

4.34 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.35 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	7
max	7
min	7

4.36 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>__>VendorId>__<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	LLCE

4.37 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43



Chapter 5

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5.1 Software Specification

Here is a list of all modules:

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Chapter 6

Module Documentation

6.1 LIN_LLCE

6.1.1 Detailed Description

Macros

- `#define LIN_43_LLCE_UNINIT`
LIN driver states.
- `#define LIN_43_LLCE_INIT`
LIN driver states.

Function Reference

- `Std_ReturnType Lin_43_LLCE_CheckWakeup (uint8 Channel)`
Validates for upper layers the wake up of LIN channel.
- `void Lin_43_LLCE_Init (const Lin_43_LLCE_ConfigType *Config)`
Initializes the LIN module.
- `Std_ReturnType Lin_43_LLCE_GoToSleepInternal (uint8 Channel)`
Put a Lin channel in the internal sleep state.
- `Std_ReturnType Lin_43_LLCE_Wakeup (uint8 Channel)`
Generates a wake up pulse.
- `Std_ReturnType Lin_43_LLCE_WakeupInternal (uint8 Channel)`
Wake up the LIN channel.

6.1.2 Macro Definition Documentation

6.1.2.1 LIN_43_LLCE_UNINIT

```
#define LIN_43_LLCE_UNINIT
```

LIN driver states.

The state LIN_43_LLCE_UNINIT means that the Lin module has not been initialized yet and cannot be used.

Definition at line 175 of file Lin_43_LLCE.h.

6.1.2.2 LIN_43_LLCE_INIT

```
#define LIN_43_LLCE_INIT
```

LIN driver states.

The LIN_43_LLCE_INIT state indicates that the LIN driver has been initialized, making each available channel ready for service.

Definition at line 183 of file Lin_43_LLCE.h.

6.1.3 Function Reference

6.1.3.1 Lin_43_LLCE_CheckWakeup()

```
Std_ReturnType Lin_43_LLCE_CheckWakeup (
    uint8 Channel )
```

Validates for upper layers the wake up of LIN channel.

This function identifies if the addressed LIN channel has been woken up by the LIN bus transceiver. This function calls EcuM and LinIf only the first time after the wake up event.

Return values

<i>E_OK</i>	No error has occurred during execution of the API
<i>E_NOT_OK</i>	An error has occurred during execution of the API

Note

Autosar Service ID: 0x0A.

6.1.3.2 Lin_43_LLCE_Init()

```
void Lin_43_LLCE_Init (
    const Lin_43_LLCE_ConfigType * Config )
```

Initializes the LIN module.

This function performs software initialization of LIN driver:

- Clears the shadow buffer of all available Lin channels
- Set LIN channel state machine of all available Lin channels to LIN_CH_OPERATIONAL
- Set frame operation state machine of all available LIN channels to LIN_CH_READY_STATE
- Set driver state machine to LIN_43_LLCE_INIT.

Parameters

in	<i>Config</i>	Pointer to LIN driver configuration set.
----	---------------	--

Returns

void

Note

Autosar Service ID: 0x00.

Synchronous, non reentrant function.

6.1.3.3 Lin_43_LLCE_GoToSleepInternal()

```
Std_ReturnType Lin_43_LLCE_GoToSleepInternal (
    uint8 Channel )
```

Put a Lin channel in the internal sleep state.

Stops any ongoing transmission, sets the channel state to LIN_CH_SLEEP and put the LIN hardware unit to a reduced power operation mode.

Parameters

in	<i>Channel</i>	LIN channel to be addressed.
----	----------------	------------------------------

Returns

Std_ReturnType.

Return values

<i>E_NOT_OK</i>	If the LIN Channel is not valid or LIN driver is not initialized or LIN Channel is in sleep state or a timeout occurs.
<i>E_OK</i>	Otherwise.

Note

Autosar Service ID: 0x09.

Synchronous, non reentrant function.

6.1.3.4 Lin_43_LLCE_Wakeup()

```
Std_ReturnType Lin_43_LLCE_Wakeup (
    uint8 Channel )
```

Generates a wake up pulse.

This function shall sent a wake up signal to the LIN bus and put the LIN channel in LIN_CH_OPERATIONAL state.

Parameters

in	<i>Channel</i>	LIN channel to be addressed.
----	----------------	------------------------------

Returns

Std_ReturnType.

Return values

<i>E_NOT_OK</i>	If the LIN driver is not in sleep state or LIN Channel is not valid or LIN driver is not initialized.
<i>E_OK</i>	Otherwise.

Note

Autosar Service ID: 0x07.

Synchronous, non reentrant function.

Module Documentation

6.1.3.5 Lin_43_LLCE_WakeupInternal()

```
Std_ReturnType Lin_43_LLCE_WakeupInternal (
    uint8 Channel )
```

Wake up the LIN channel.

This function shall put the LIN channel in LIN_CH_OPERATIONAL state without sending a wake up signal to the LIN bus

Parameters

in	Channel	LIN channel to be addressed.
----	---------	------------------------------

Returns

Std_ReturnType.

Return values

<i>E_NOT_OK</i>	If the LIN driver is not in sleep state or LIN Channel is not valid or LIN driver is not initialized.
<i>E_OK</i>	Otherwise.

Note

- Autosar Service ID: 0x0B.
- Asynchronous, non reentrant function.

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